Area Between Curves

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Area between two curves**

Find the area of the region bounded by $y = x^2$ and $y = x + 2$.

Intersections: $x^2 = x + 2$, so $x^2 - x - 2 = 0$, $(x - 2)(x + 1) = 0$: $x = -1, 2$. Area $= \int_{-1}^{2} [(x + 2) - x^2]\,dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2} = \frac{9}{2}$.

2 **[Foundational]** • **Target: Area between two curves**

The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is shown. Find its area.

$\int_0^4 \sqrt{x}\,dx = \left[\frac{2}{3}x^{3/2}\right]_0^4 = \frac{16}{3}$.

3 **[Foundational]** • **Target: Setting up with respect to y, and total area**

Find the area enclosed by $x = y^2$ and $x = 2y$ by integrating with respect to y.

Intersections: $y^2 = 2y$ gives $y = 0, 2$. Area $= \int_0^2 (2y - y^2)\,dy = \left[y^2 - \frac{y^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

4 **[Foundational]** • **Target: Setting up with respect to y, and total area**

Find the total area between $f(x) = \sin x$ and the x-axis on $[0, 2\pi]$ (note the curve is below the axis on part of this interval).

By symmetry, $\int_0^{\pi} \sin x\,dx = 2$ and $\left|\int_{\pi}^{2\pi} \sin x\,dx\right| = 2$. Total area $= 2 + 2 = 4$.

5 **[Foundational]** • **Target: More area problems**

Find the area of the region bounded by $y = x^2 - 4$ and $y = 4 - x^2$.

Intersections: $x^2 - 4 = 4 - x^2$, so $2x^2 = 8$, $x = \pm 2$. Upper curve is $y = 4 - x^2$: Area $= \int_{-2}^{2} [(4 - x^2) - (x^2 - 4)]\,dx = \int_{-2}^{2} (8 - 2x^2)\,dx = \left[8x - \frac{2x^3}{3}\right]_{-2}^{2} = \frac{64}{3}$.

6 **[Foundational]** • **Target: More area problems**

Find the area between $y = x^3$ and $y = x$ on $[0, 1]$ (note that $y = x$ is above $y = x^3$ on this interval).

Area $= \int_0^1 (x - x^3)\,dx = \left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.

7 **[Foundational]** • **Target: Areas requiring case splits**

Find the area between $y = x^3$ and $y = x$ on $[-1, 1]$ (note the curves cross at $x = 0$, so the identity of the upper curve switches).

On $[-1, 0]$, $x^3 \geq x$; on $[0, 1]$, $x \geq x^3$. Area $= \int_{-1}^{0} (x^3 - x)\,dx + \int_0^1 (x - x^3)\,dx$. By symmetry both integrals equal $\frac{1}{4}$ (using the earlier result), so total area $= \frac{1}{2}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Areas requiring case splits**

Find the area enclosed between $y = 2x$ and $y = x^2$ on the interval where $y = 2x$ is above $y = x^2$.

Intersections: $2x = x^2$ gives $x = 0, 2$. On $(0, 2)$, $2x > x^2$: Area $= \int_0^2 (2x - x^2)\,dx = \left[x^2 - \frac{x^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

9 **[Intermediate]** • **Target: Areas involving trigonometric and exponential curves**

Find the area between $y = \cos x$ and $y = 0$ on $\left[0, \frac{\pi}{2}\right]$.

Area $= \int_0^{\pi/2} \cos x\,dx = [\sin x]_0^{\pi/2} = 1 - 0 = 1$.

10 **[Intermediate]** • **Target: Areas involving trigonometric and exponential curves**

Find the area between $y = e^x$ and $y = e^{-x}$ on $[0, 1]$ (note $e^x \geq e^{-x}$ on this interval).

Area $= \int_0^1 (e^x - e^{-x})\,dx = [e^x + e^{-x}]_0^1 = (e + e^{-1}) - (1 + 1) = e + \frac{1}{e} - 2$.

11 **[Intermediate]** • **Target: Visualizing a region bounded by two curves**

The graphs of $f(x) = x^2 - 4$ and $g(x) = 4 - x^2$ from an earlier problem are shown together. Confirm from the picture that the region is symmetric about the y-axis, and use this to verify the earlier area answer $\frac{64}{3}$ using $2\int_0^2 (8 - 2x^2)\,dx$.

The region is symmetric about $x = 0$ since both curves are even functions. $2\int_0^2 (8 - 2x^2)\,dx = 2\left[8x - \frac{2x^3}{3}\right]_0^2 = 2\left(16 - \frac{16}{3}\right) = 2 \cdot \frac{32}{3} = \frac{64}{3}$, matching the earlier full-interval computation.

12 **[Intermediate]** • **Target: Area between a curve and a horizontal line**

Find the area between $y = 6 - x^2$ and $y = 2$.

Intersections: $6 - x^2 = 2$ gives $x = \pm 2$. Area $= \int_{-2}^2 [(6 - x^2) - 2]\,dx = \int_{-2}^2 (4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^2 = 2\left(8 - \frac{8}{3}\right) = \frac{32}{3}$.

13 **[Intermediate]** • **Target: Area between a curve and a horizontal line**

Find the area between $y = \sqrt{x}$ and $y = \frac{x}{2}$ for $x \geq 0$.

Intersections: $\sqrt{x} = \frac{x}{2}$ gives $x = 0$ or $x = 4$. On $(0, 4)$, $\sqrt{x} \geq \frac{x}{2}$: Area $= \int_0^4 \left(\sqrt{x} - \frac{x}{2}\right)dx = \left[\frac{2}{3}x^{3/2} - \frac{x^2}{4}\right]_0^4 = \frac{16}{3} - 4 = \frac{4}{3}$.

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14 [Intermediate] • Target: Area between curves in applied contexts

Two runners' velocities (m/s) over a 10-second race are $v_1(t) = 2t$ and $v_2(t) = t + 3$. The area between these curves represents the gap in distance traveled. Find this gap over $[0, 10]$.

Intersection: $2t = t + 3$ gives $t = 3$. For $t < 3$, $v_2 > v_1$; for $t > 3$, $v_1 > v_2$. Gap $= \left|\int_0^3 (v_2 - v_1)dt\right| + \left|\int_3^{10} (v_1 - v_2)dt\right|$. $\int_0^3 (3 - t)\,dt = \left[3t - \frac{t^2}{2}\right]_0^3 = 9 - 4.5 = 4.5$. $\int_3^{10} (t - 3)\,dt = \left[\frac{t^2}{2} - 3t\right]_3^{10} = (50 - 30) - (4.5 - 9) = 20 + 4.5 = 24.5$. Total gap $= 4.5 + 24.5 = 29$ m.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 3.01 The Derivative from First Principles

Find the area of the region bounded by $y = x^2 - 2x$ and the x-axis on $[0, 2]$ (note the curve dips below the axis on this interval).

$y = x^2 - 2x = x(x - 2)$ is ≤ 0 on $[0, 2]$. Area $= \left|\int_0^2 (x^2 - 2x)\,dx\right| = \left|\left[\frac{x^3}{3} - x^2\right]_0^2\right| = \left|\frac{8}{3} - 4\right| = \frac{4}{3}$.

16 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Find the area between $y = x^3 - x$ and the x-axis on $[-1, 1]$, splitting at the roots $x = -1, 0, 1$ where the sign changes.

On $[-1, 0]$, $x^3 - x \ge 0$; on $[0, 1]$, $x^3 - x \le 0$. Area $= \int_{-1}^0 (x^3 - x)\,dx + \left|\int_0^1 (x^3 - x)\,dx\right| = \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_{-1}^0 + \left|\left[\frac{x^4}{4} - \frac{x^2}{2}\right]_0^1\right| = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

17 [Challenging] • Target: Areas involving trigonometric and exponential curves • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Region bounded by $y = 3 - x^2$ and $y = -1$. Area:.

Final Answer: 32/3. Intersect at $x = \pm 2$. $\int_{-2}^2 (4 - x^2)dx = 2\int_0^2 = 2[4x - x^3/3]_0^2 = 2(8 - 8/3) = 32/3$.

18 [Challenging] • Target: Visualizing a region bounded by two curves • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Volume of region bounded by $x^2/4 + y^2/9 = 1$ rotated about the x-axis:.

Final Answer: 24π. Rotating $\frac{x^2}{4} + \frac{y^2}{9} = 1$ about the x-axis gives $V = \pi\int_{-2}^2 9\left(1 - \frac{x^2}{4}\right)dx = 9\pi\left[x - \frac{x^3}{12}\right]_{-2}^2 = 24\pi$.

19 [Challenging] • Target: Area between a curve and a horizontal line • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Find the total area enclosed by the cardioid $r = 2(1 - \cos\theta)$.

Final Answer: 6π. The polar area is $A = \frac{1}{2}\int_0^{2\pi} 4(1 - 2\cos\theta + \cos^2\theta)\,d\theta = 2(2\pi + \pi) = 6\pi$.

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20 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 6.01 Related Rates

Centroid $\bar{y}$ of the region bounded by $y = x^2$ and $y = 4$:.

Final Answer: $\bar{y} = 12/5$. $A = 32/3$; $M_x = \int_{-2}^{2} \frac{1}{2}(16 - x^4)dx = 128/5$; $\bar{y} = (128/5)/(32/3) = 12/5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vertical Slices (Integration with Respect to x)

Find the area enclosed by $y = x^2$ and $y = 2x - x^2$.

a. 1/3 ✓ **b.** 2/3

c. 1/6 **d.** 1

Intersections: $x^2 = 2x - x^2 \Rightarrow 2x^2 - 2x = 0 \Rightarrow x = 0, 1$. Top curve is $2x - x^2$. Area $= \int_0^1 (2x - 2x^2)\,dx = [x^2 - \frac{2}{3}x^3]_0^1 = 1 - 2/3 = 1/3$.

22 [Foundational] • Target: Vertical Slices (Integration with Respect to x)

Find the area between $y = x^2$ and $y = 4$.

a. 32/3 ✓ **b.** 16/3

c. 8 **d.** 64/3

Intersections at $x = \pm 2$. By symmetry: $2\int_0^2 (4 - x^2)\,dx = 2[4x - x^3/3]_0^2 = 2[8 - 8/3] = 2(16/3) = 32/3$.

23 [Foundational] • Target: Vertical Slices (Integration with Respect to x)

Find the area between $y = \sqrt{x}$ and $y = x^2$.

a. 1/3 ✓ **b.** 2/3

c. 1/6 **d.** 1/2

Intersections at $x = 0$ and $x = 1$. $\int_0^1 (x^{1/2} - x^2)\,dx = [\frac{2}{3}x^{3/2} - \frac{x^3}{3}]_0^1 = 2/3 - 1/3 = 1/3$.

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24 [Foundational] • Target: Horizontal Slices (Integration with Respect to y)

Find the area enclosed by $x = y^2$ and $x = y + 2$.

a. 9/2 ✓

b. 16/3

c. 7/2

d. 9

Intersections: $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y-2)(y+1) = 0 \Rightarrow y = -1, 2$. Area $= \int_{-1}^{2} (y + 2 - y^2)\,dy = [\frac{y^2}{2} + 2y - \frac{y^3}{3}]_{-1}^{2} = (2 + 4 - 8/3) - (1/2 - 2 + 1/3) = 10/3 - (-7/6) = 27/6 = 9/2$.

25 [Foundational] • Target: Horizontal Slices (Integration with Respect to y)

Why is integrating with respect to y advantageous for curves like $x = y^2 - 4$ and $x = 0$?

a. **The right curve minus left curve is a single continuous expression without piecewise splitting** ✓

b. Integration with respect to x is impossible

c. It changes the area value

d. The bounds are always 0 and 1

With horizontal slices, the right boundary is $x = 0$ and the left boundary is $x = y^2 - 4$ across the entire range $y \in [-2, 2]$, avoiding splitting into upper and lower halves.

26 [Foundational] • Target: Horizontal Slices (Integration with Respect to y)

Find the area enclosed between $x = 4 - y^2$ and the y-axis.

a. 32/3 ✓

b. 16/3

c. 8

d. 16

Intersections with y-axis ($x = 0$): $y = \pm 2$. Area $= \int_{-2}^{2} (4 - y^2)\,dy = 2[4y - y^3/3]_0^2 = 2[8 - 8/3] = 32/3$.

27 [Foundational] • Target: Crossing Curves and Multiple Regions

Find the total area enclosed between $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \pi/2$.

a. $2\sqrt{2} - 2$ ✓

b. $\sqrt{2} - 1$

c. 2

d. 1

Intersection at $x = \pi/4$. On $[0, \pi/4]$, $\cos x \geq \sin x$; on $[\pi/4, \pi/2]$, $\sin x \geq \cos x$.
$\int_0^{\pi/4} (\cos x - \sin x)dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)dx = (\sqrt{2} - 1) + (\sqrt{2} - 1) = 2\sqrt{2} - 2$.

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28 [Foundational] • Target: Crossing Curves and Multiple Regions

Find the area enclosed between $y = x^3 - x$ and $y = 0$ for $x \in [-1, 1]$.

a. $1/2$ ✓

b. 0

c. $1/4$

d. 1

Due to odd symmetry, integrating directly gives 0 (signed area). Absolute area $= 2\int_0^1 -(x^3 - x)\,dx = 2[\frac{x^2}{2} - \frac{x^4}{4}]_0^1 = 2(1/2 - 1/4) = 2(1/4) = 1/2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Crossing Curves and Multiple Regions

Find the area enclosed between $y = x^3$ and $y = 4x$.

a. 8 ✓

b. 4

c. 16

d. 0

Intersections: $x^3 = 4x \Rightarrow x(x-2)(x+2) = 0 \Rightarrow x = -2, 0, 2$. By symmetry, total area $= 2\int_0^2 (4x - x^3)\,dx = 2[2x^2 - x^4/4]_0^2 = 2[8 - 4] = 2(4) = 8$.

30 [Intermediate] • Target: Applications and Boundaries

Find the area of the region bounded by $y = e^x$, $y = e^{-x}$, and $x = 1$.

a. $e + 1/e - 2$ ✓

b. $e - 1/e$

c. $e + 1/e$

d. $e - 2$

On $[0, 1]$, $e^x \geq e^{-x}$. Area $= \int_0^1 (e^x - e^{-x})\,dx = [e^x + e^{-x}]_0^1 = (e + e^{-1}) - (1 + 1) = e + 1/e - 2$.

31 [Intermediate] • Target: Applications and Boundaries

Find the area bounded by $y = \frac{1}{x}$, the x-axis, $x = 1$, and $x = e^2$.

a. 2 ✓

b. 1

c. e^2

d. $2e$

$\int_1^{e^2} \frac{1}{x}\,dx = [\ln x]_1^{e^2} = \ln(e^2) - \ln(1) = 2 - 0 = 2$.

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32 [Intermediate] • Target: Applications and Boundaries

The Gini index measures income inequality as twice the area between the line of perfect equality $y = x$ and the Lorenz curve $L(x) = x^2$ on $[0, 1]$. Find the Gini index.

a. $1/3$ ✓

b. $1/6$

c. $2/3$

d. $1/2$

Area $= \int_0^1 (x - x^2)\,dx = [x^2/2 - x^3/3]_0^1 = 1/2 - 1/3 = 1/6$. Gini index $= 2 \times 1/6 = 1/3$.

33 [Intermediate] • Target: Area between two curves

Area of the region enclosed by the lemniscate $r^2 = 4\cos 2\theta$:

a. 4 ✓

b. 2

c. 8

d. π

$A = \frac{1}{2}\int_{-\pi/4}^{\pi/4} 4\cos 2\theta \cdot 2\,d\theta = 4$.

34 [Intermediate] • Target: Setting up with respect to y, and total area

Area of region bounded by parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (full astroid):

a. $3\pi/8$ ✓

b. $3\pi/4$

c. $\pi/4$

d. $\pi/2$

Astroid area $= 3\pi a^2/8$ with $a = 1$: $3\pi/8$.

35 [Intermediate] • Target: More area problems

Tangent line to parametric curve is horizontal when:

a. $dx/dt = 0$

b. $dy/dt = 0$ **and** $dx/dte0$ ✓

c. $t = 0$

d. $dy/dt = dx/dt$

Applying Parametric and Polar principles, Horizontal tangent: $dy/dt = 0$. Therefore, the verified correct option is $dy/dt = 0$ and $dx/dte0$.

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36 [Intermediate] • Target: Areas requiring case splits

Tangent line to parametric curve is vertical when:

a. $dy/dt = 0$

b. $dx/dt = 0$ **and** $dy/dte0$ ✓

c. $dx/dt = dy/dt$

d. $t = 0$

Applying Parametric and Polar principles, Vertical tangent: $dx/dt = 0$. Therefore, the verified correct option is $dx/dt = 0$ and $dy/dte0$.

37 [Intermediate] • Target: Areas involving trigonometric and exponential curves

Pappus's theorem: volume of solid of revolution = $2\pi\bar{x} \cdot A$ where $\bar{x}$ is the centroid of the region and A is the:

a. arc length

b. surface area

c. **area of the region** ✓

d. cross-sectional area

Applying Applications of Integration principles, Pappus's centroid theorem. Therefore, the verified correct option is area of the region.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Visualizing a region bounded by two curves • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Surface area of cone with radius r, slant height l:

a. $\pi r^2 l$

b. $\pi r l$ ✓

c. $2\pi r l$

d. $\pi r(r + l)$

Applying Applications of Integration principles, Lateral surface area $= \pi r l$. Therefore, the verified correct option is $\pi r l$.

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39 [Challenging] • Target: Area between a curve and a horizontal line • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Centroid of semicircular region radius R, $y \geq 0$:

a. $\bar{y} = R/2$

b. $\bar{y} = 4R/(3\pi)$ ✓

c. $\bar{y} = 2R/3$

d. $\bar{y} = R/\pi$

Applying Applications of Integration principles, $\bar{y} = 4R/(3\pi)$. Therefore, the verified correct option is $\bar{y} = 4R/(3\pi)$.

40 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 6.01 Related Rates

Fluid force: pressure at depth d is $\rho g d$. Force on area A:

a. $\rho g A$

b. ρA^2

c. $\rho g d A$ ✓

d. $g d$

Applying Applications of Integration principles, $F = PA = \rho g d A$. Therefore, the verified correct option is $\rho g d A$.

41 [Challenging] • Target: Vertical Slices (Integration with Respect to x) • Cross-Lesson: 6.02 Optimization and Curve Sketching

Area under one arch of $y = \sin x$ (from 0 to π):

a. 0

b. 2 ✓

c. 1

d. π

Applying Applications of Integration principles, $\int_0^{\pi} \sin x \, dx = 2$. Therefore, the verified correct option is 2.

42 [Challenging] • Target: Horizontal Slices (Integration with Respect to y) • Cross-Lesson: 7.01 Integration by Substitution

Area of astroid $x = \cos^3 t$, $y = \sin^3 t$:

a. $3\pi/4$

b. $3\pi/8$ ✓

c. $\pi/2$

d. $3\pi/2$

Applying Applications of Integration principles, $A = \frac{3\pi}{8}$. Therefore, the verified correct option is $3\pi/8$.

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43 [Challenging] • Target: Crossing Curves and Multiple Regions • Cross-Lesson: 7.02 Integration by Parts

Total area between $y = x$ and x-axis on $[-2, 2]$:

a. 0

b. 2

c. 4 ✓

d. 8

Applying Fundamental Theorem principles, $2\int_0^2 x\,dx = 2 \cdot 2 = 4$. Therefore, the verified correct option is 4 .

44 [Challenging] • Target: Applications and Boundaries • Cross-Lesson: 8.02 Volumes of Revolution

Find the area of the surface formed by revolving $y = \sqrt{x}$, $0 \le x \le 3$, about the x-axis.

a. $\frac{\pi(-1+13\sqrt{13})}{6}$ ✓

b. $\frac{\pi(-1+13\sqrt{13})}{12}$

c. $4\sqrt{3}\pi$

d. $\frac{9\pi}{2}$

$S = 2\pi\int_0^3 \sqrt{x}\sqrt{1+\frac{1}{4x}}\,dx = \frac{\pi(-1+13\sqrt{13})}{6}$.

45 [Challenging] • Target: Area between two curves • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Find the area of the surface formed by revolving $y = \sqrt{x}$, $0 \le x \le 1$, about the x-axis.

a. $\frac{\pi(-1+5\sqrt{5})}{6}$ ✓

b. $\frac{\pi(-1+5\sqrt{5})}{12}$

c. $\frac{4\pi}{3}$

d. $\frac{\pi}{2}$

$S = 2\pi\int_0^1 \sqrt{x}\sqrt{1+\frac{1}{4x}}\,dx = \frac{\pi(-1+5\sqrt{5})}{6}$.

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